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Primer

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Atopic dermatitis

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Abstract

Atopic dermatitis (AD) is the most common inflammatory skin disease and carries the highest disability-adjusted life-years burden, ranking 15th among all non-fatal diseases globally. It is characterized by intensely itchy skin and is associated with multiple comorbidities, such as food allergy, asthma, allergic rhinitis and eosinophilic oesophagitis, which are mainly driven by type 2 immune responses. Other comorbidities include mental health disorders, disordered bone health, and cutaneous and extracutaneous infections. AD is also associated with other immune-mediated inflammatory diseases, including alopecia areata, vitiligo and inflammatory bowel disease. AD most often starts in the first 2 years of life but can occur at any life stage and onset at >60 years of age is increasingly common. The twenty-first century has brought greater insights into disease pathology, with an understanding of the complex interplay between the skin barrier, cutaneous and systemic immune pathways, cutaneous microbiome and neural networks. This improved mechanistic understanding has enabled rational drug design and a shift from non-specific broad immunomodulation to targeted biologic therapies and small molecules for severe disease and from topical corticosteroids to next-generation therapies for mild and moderate disease. Yet, considerable global inequity remains in access to these novel therapeutics.

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Introduction

Atopic dermatitis (AD), also referred to as atopic eczema, is a chronic inflammatory skin disease defined clinically by recurrent eczematous lesions and intense pruritus. Typical lesions are poorly demarcated erythematous patches and plaques that may ooze, crust or vesiculate (form small fluid-filled blisters) in acute disease and evolve to scaling, fissuring and lichenification, a visible thickening of the skin caused by chronic rubbing and scratching, in more chronic disease[1]. The onset of AD can begin at any age but most commonly starts in early childhood, with an age-dependent distribution of lesions[2]. Disease trajectories range from transient childhood AD to persistent, recurrent disease across the life course, and late-onset presentations are increasingly recognized in older adults[3,4]. AD contributes substantially to the global burden of skin disease and is associated with marked impairment in quality of life, driven by persistent itch, sleep disturbance and reduced daily functioning[5,6].

AD is best viewed as an umbrella diagnosis that encompasses marked clinical and biological heterogeneity. The disease varies in age at onset, anatomical distribution, morphology and course, and shows substantial differences in trigger susceptibility and treatment response; this heterogeneity underpins the wide range in long-term outcomes and the frequent need for individualized management. Common clinical patterns include flexural disease (eczema predominantly in body folds, such as the elbows and knees) as well as variants such as head-and-neck involvement, hand-predominant disease, nummular eczema (coin-shaped patches of skin inflammation), follicular accentuation (eczema that is particularly prominent around hair follicles) and prurigo nodularis-like disease (with very itchy, firm skin nodules caused by chronic scratching)[4,7,8].

Current conceptual frameworks of AD pathogenesis centre on reciprocal interactions between epidermal barrier impairment, immune dysregulation, skin dysbiosis and neuroimmune pathways that sustain itch and the itch–scratch cycle. Genetic susceptibility for AD includes variants affecting barrier structure and function, exemplified by loss-of-function (LoF) variants in *FLG* encoding filaggrin; risk is further shaped by environmental exposures[9–12]. In many individuals, inflammatory circuits are dominated by type 2 cytokines, such as IL-4 and IL-13, with downstream effects on epidermal differentiation, lipid composition, antimicrobial defence and pruritus pathways[13,14]. Microbial dysbiosis, particularly an increased *Staphylococcus aureus* burden, is associated with disease activity and can exacerbate barrier dysfunction, inflammation and itch[15–20]. Neuroimmune signalling links cutaneous inflammation to sensory neurons, amplifying pruritus and scratching, thereby perpetuating disease activity[21,22].

AD is frequently associated with IgE sensitization and with other type 2-associated conditions, such as food allergy, asthma and allergic rhinitis, although these comorbidities are not universal[23]. Throughout this Primer, 'AD' denotes the disease entity, whereas 'eczema' refers to lesion morphology. 'Atopy' describes a predisposition to IgE sensitization and allergic disease (present in some but not all individuals), and 'type 2 inflammation' denotes immune circuits dominated by cytokines, such as IL-4, IL-13 and IL-31 (Fig. 1); type 2 circuits are central to many presentations of AD but do not fully explain all phenotypes.

No single classification captures AD in its entirety, and clinical and mechanistic schemes are therefore used in parallel. Clinical frameworks organize AD by age-related distribution of lesions, by morphology and site, and by multidimensional severity[4,7,8]. Mechanistic frameworks have proposed candidate immune endotypes, largely defined by the relative contributions of inflammatory axes and their associated molecular signatures[24–26]; these concepts are increasingly relevant to targeted therapy and to emerging approaches that distinguish disease control, remission and disease modification (Box 1).

In this Primer, we summarize the epidemiology of AD, including known risk factors, discuss current knowledge of disease pathophysiology, highlighting the interplay between the skin barrier, cutaneous and systemic immune pathways, the cutaneous microbiome and neural networks, and discuss how phenotype and endotype concepts can be integrated to support diagnosis, assessment and management.

Epidemiology

Prevalence

AD affects up to 20% of children and 10% of adults worldwide. AD ranks 15th among non-fatal diseases and first among all skin diseases, as measured by disability-adjusted life-years, with considerable geographical variation[5]. AD is most prevalent among children and young people, with >75% of affected individuals presenting within the first 5 years of life[27]. Population-based studies have shown a bimodal age distribution, with a primary peak in childhood in which prevalence is reported to be as high as 20%[28] and a secondary, smaller peak in individuals >60 years of age, in whom prevalence is reported at ~10%[29].

Geographical prevalence

The highest 1-year period prevalence has been reported among children in central sub-Saharan Africa at 5.2%[30]. High 1-year physician- or dermatologist-diagnosed period prevalences in children were also observed in Latin America (Andean 4.93% and Central 4.75%), southern sub-Saharan Africa (4.64%), and high-income North America (5.01%). By contrast, the lowest 1-year prevalence in children was observed in Central Asia (2.4%), closely followed by Eastern and Central Europe (2.98%)[30]. Using age-standardized disability-adjusted life-years, high-income Asia Pacific and Western Europe were shown to have the highest AD burden after Andean Latin America and high-income North America[5].

Although AD is not only a disease of affluent settings in Northern Europe and North America, it is generally commoner among populations that lead a 'Western' lifestyle and move from rural into urban settings, highlighting the important environmental contribution to disease development. This has been demonstrated in the largest epidemiological study of AD prevalences, the International Study of Asthma and Allergies in Childhood, which recruited >1.3 million children from >200 centres in almost 100 countries, using the same validated methodology across all sites[31–33] (Fig. 2). Although these estimates provide valuable insight into the global burden of AD, there remain considerable gaps in available epidemiological data, making estimates uncertain in some regions[30,33,34] (Supplementary Fig. 1).

Increased disease burden is linked to high-income, urbanized nations, suggesting a link to environmental and lifestyle factors, including a reduced exposure to helminth parasites and farm animals, and air pollution[5,35]. For instance, direct contact with farm animals during the last trimester of pregnancy and in early life conveys a protective effect against allergic tissue inflammation, including AD, in the offspring. Chronic helminth infections, where endemic, also actively reduce T helper 2 (T_H2) cell inflammatory circuits, whereas air pollutants, such as volatile organic and nitrogen compounds, damage skin barrier stratum corneum lipids and proteins, contributing to eczematous skin inflammation and the risk of allergic sensitization through the skin, for instance to house dust mite[36]. The far-reaching threats of climate change on public health extend to the disease course and management of AD. For instance, the rise in temperature and pollutant concentrations

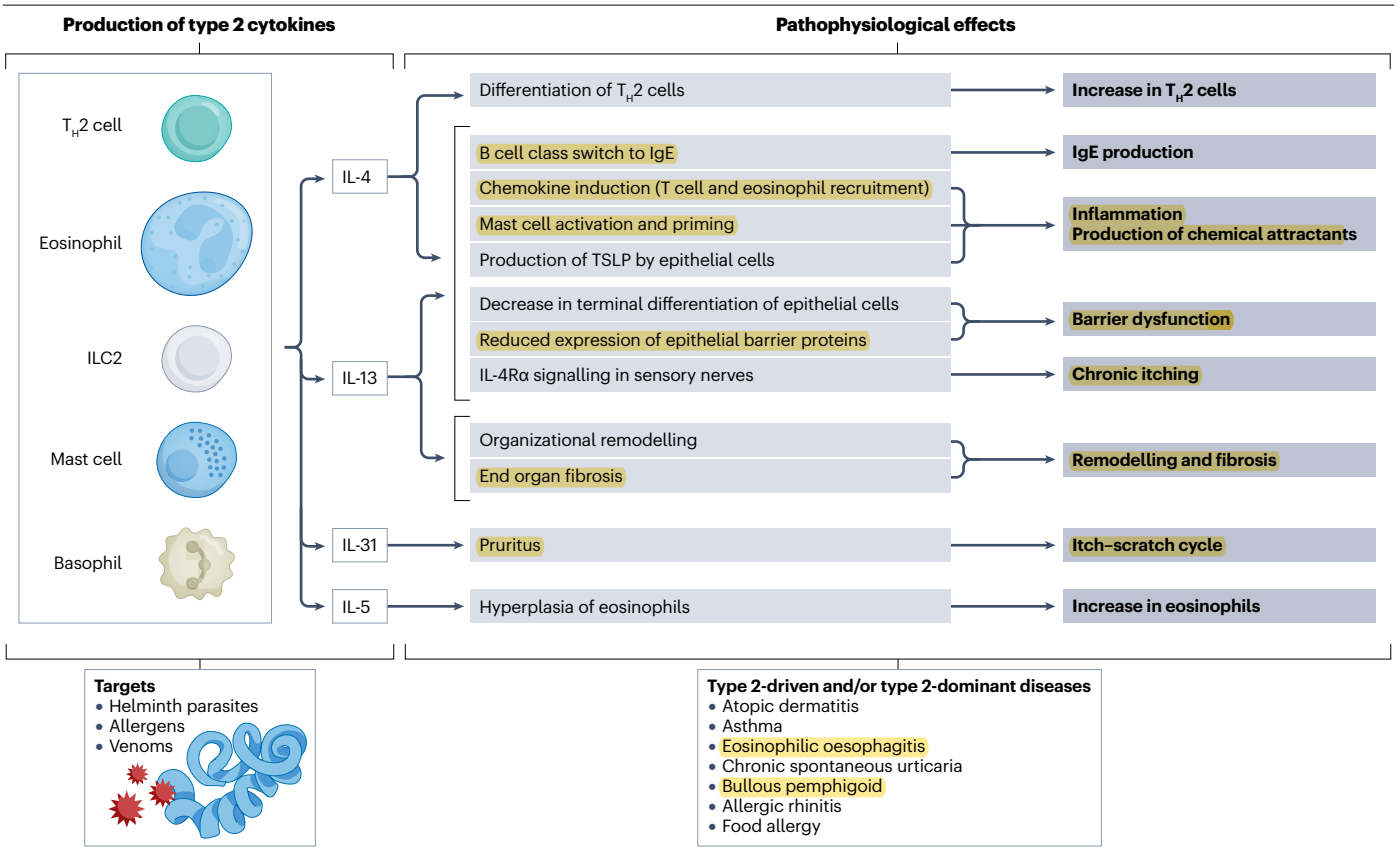


Fig. 1 | Type 2 immunity and related diseases. Type 2 immunity has a physiological role in host defence against parasites, venoms and allergens but dysregulated type 2 immunity is a key driver of many highly prevalent atopic diseases. ILC2, type 2 innate lymphoid cell; T$_H$2 cell, T helper 2 cell.

can cause an increase in the concentrations and allergenicity of plant pollens; therefore, individuals with AD who are sensitized to trees and grasses often experience flares during the pollen season in air-exposed areas[37]. Extreme temperatures, Ultraviolet B irradiation and toxins from traffic-related air pollution can disrupt the skin barrier and exacerbate AD severity[38]. Reports on the effect of climate factors on the prevalence and severity of AD are highly variable, with some studies showing a positive relationship between increasing temperatures and AD prevalences, and others associating higher temperatures with poor symptomatic control[39,40]. In summary, the considerable variation in the burden of AD at the population level across geographical settings is in part driven by an interplay of environmental factors that either promote cutaneous inflammatory circuits directly or enhance sensitization to environmental allergens that can exacerbate AD skin.

Comorbidities

The close relationship between AD and other atopic type 2 immune-mediated conditions (food and respiratory allergies in particular) is well established. It is reported that among people with AD, ~40% also have rhinitis, ~26% have asthma and ~14% have both asthma and rhinitis[23]. This is in part explained by genetic traits linking these diseases, including inheritance of loss-of-function mutations in the filaggrin skin barrier gene[11]. There is also mounting evidence that both food and respiratory allergies develop through transcutaneous sensitization, especially in the presence of eczematous skin[41,42]. This would also explain why almost all children react to foods on first oral exposure[43,44].

There is also growing recognition of non-type 2 comorbidities. These include cardiometabolic and psychiatric conditions, such as stroke, myocardial infection, attention deficit hyperactivity disorder, anxiety, depression and suicidal ideation[23]. However, the association between AD and cardiovascular morbidity is much less established than the risk of cardiovascular disease in people with psoriasis[23]. AD has also been associated with secondary cutaneous infections with viruses, including herpes simplex virus (HSV) and molluscum contagiosum, bacterial infections, including impetigo, and yeast infections, including *Malassezia* species infection[45,46]. Bacterial infections beyond the skin, including osteomyelitis, are more common in people with AD than in the general population, as is herpes zoster[47]. Prurigo nodularis is also observed more commonly in people with AD, and is often difficult to distinguish as a distinct disease state in people with concurrent AD[48].

Mechanisms/pathophysiology

AD is a clinically defined disease with substantial heterogeneity across age groups, ethnicities, disease trajectories and comorbidity patterns, and it is likely that multiple endotypes (disease subtypes defined by pathophysiological mechanism) are currently grouped under the umbrella term AD. Clinically, AD is often stratified into 'extrinsic' and 'intrinsic' forms (often based on the presence of IgE sensitization) or into comorbidity-associated phenotypes

Box 1 | From control to remission and modification: emerging therapeutic concepts with persistent uncertainties

Historically, clinical trials and routine care in atopic dermatitis (AD) have focused on short-term improvement in signs and symptoms, quantified by composite disease activity scores, such as Scoring Atopic Dermatitis (SCORAD), or combinations of scores for clinical signs and patient-reported outcomes, such as the Eczema Area and Severity Index (EASI) and the Peak Pruritus Numerical Rating Scale, which represent the most immediate and easily attainable treatment targets[254,298].

Recognizing that patients value long-term disease control more than point-in-time improvements, the Harmonizing Outcome Measures for Eczema (HOME) initiative subsequently identified long-term disease control (acceptable symptom control over time with a manageable treatment burden) as a core outcome domain in AD and stimulated the development of validated patient-reported control measures, notably the Atopic Dermatitis Control Tool and Recap of Atopic Eczema, which capture day-to-day fluctuations, treatment burden and impact on life[249,299,300], but are not yet widely recognized. Building on treat-to-target strategies in rheumatoid arthritis, in which sustained remission is now the principal goal[301], remission in AD has been proposed as a more ambitious outcome beyond disease control, encompassing sustained absence of clinically relevant signs and symptoms over extended periods, either with ongoing therapy or after treatment discontinuation. Although a first international consensus-based framework has now proposed definitions of "low disease activity", "very low disease activity", "on-drug complete control" and "off-drug remission" based on combined clinician-reported signs and patient-reported symptoms[302], a universally accepted definition of remission in AD is still lacking, and the practical applicability and uptake of these definitions remain to be determined[303–306].

In parallel, disease modification in AD has been proposed as durable, therapy-independent remission accompanied by absent or minimal biological disease activity and a potentially altered long-term disease trajectory. Although the natural course of AD remains incompletely understood, this concept is often linked to the possibility of reducing long-term inflammatory burden and also associated type 2 comorbidities, and may represent the ultimate therapeutic ambition. This state may be most achievable through early and appropriate intervention targeting type 2 inflammation[253,304,307]. Recent work has, therefore, focused on defining disease activity indices that integrate clinician-reported scores, patient-reported outcomes, such as Atopic Dermatitis Control Tool and Recap of Atopic Eczema, and candidate biomarkers for future trials addressing remission and disease modification[253,308]. However, prospective studies incorporating early intervention designs, therapy withdrawal arms and long off-drug follow-up are still needed to validate operational definitions of disease control, remission and disease modification in AD and to understand whether, when, in whom and which systemic treatment may alter long-term trajectories.

(for example, AD with food allergy and/or asthma), but these categories have not consistently translated into clinically actionable endotypes[49]. Transcriptomic profiling may help identify biologically distinct subgroups with a greater contribution from non-type 2 inflammatory circuits and potentially different responses to therapy classes; however, most proposed molecular signatures remain research tools and still require independent replication and clinical validation before they can be considered broadly clinically useful[50]. As an organizing principle, we here use a quadripartite framework linking the four major pathophysiological mechanisms in AD, which are epidermal barrier dysfunction, microbial imbalance, immune dysregulation and neural sensitization.

Genetic architecture

A family history of atopic disease, especially AD, is a major risk factor for developing AD; heritability is estimated to be ~75%[51]. LoF variants in the filaggrin gene (*FLG*) remain the most robust genetic risk factor for AD. In a semi-dominant fashion, homozygous and compound heterozygous LoF variants cause ichthyosis vulgaris and a complete absence of biologically active filaggrin peptides, while heterozygotes show reductions in filaggrin of ~50% and a threefold to fivefold increased risk of AD. Around 8–10% of Europeans carry one *FLG* LoF variant[52]. Mechanistically, filaggrin deficiency leads to a disturbed aggregation of keratin intermediate filaments and reduced amounts of breakdown products (including free amino acids, pyrrolidone carboxylic acid and urocanic acid) that contribute to natural moisturization, leading to decreased skin hydration, increased transepidermal water loss, perturbed lipid lamellar organization and impaired antimicrobial defence, and promotes alarmin-driven type 2 inflammation[53].

Large-scale genome-wide association studies and sequencing efforts have expanded the genetic landscape beyond *FLG*, implicating additional genes within the epidermal differentiation complex. Outside structural barrier loci, a substantial fraction of risk maps to immunoregulatory regions such as 5q31 (*IL13–KIF3A*), 5q22 (*TSLP*) and 11q13 (*OVOL1*), with a smaller but reproducible signal in the HLA region[10]. Integrative annotations show enrichment in DNase I-hypersensitive sites and expression quantitative trait loci in immune cells and keratinocytes[10]. An expanded multi-ancestry genome-wide association study found a total of 101 risk loci with strong enrichment for T cell signalling (T_H1, T_H2, T_H17 and regulatory T) and for keratinocyte programmes in lesional skin and disease progression modules, reinforcing barrier–immune crosstalk as core disease biology[9].

Immune dysregulation

Immune dysregulation is a central element in the pathogenesis of AD, involving a tightly interwoven network of innate and adaptive immune mechanisms (Fig. 3). The disease is underpinned by type 2 inflammation driven primarily by IL-13 and IL-4, with IL-4 serving as an early instructive signal for T_H2 cell polarization and a central amplifier of humoral type 2 immunity[54], whereas IL-13 predominantly mediates tissue-level effector circuits in the skin[55]. The traditional concept of 'type 2 dominance' has been refined into a more complex framework in which early innate immune activation orchestrates, and then integrates with, adaptive responses[56]. Innate lymphoid cells, particularly group 2

innate lymphoid cells, are activated in response to epithelial-derived alarmins such as TSLP and IL-33 (and in some settings IL-25), producing abundant IL-5 and IL-13 rapidly after barrier disruption[57,58]. These cytokines, together with fibroblast-derived eotaxins (CCL11 and CCL24) initiate eosinophil recruitment, augment vascular permeability, and create a permissive environment for the activation and expansion of antigen-specific T helper cells in part via TSLP-conditioned dendritic cells expressing OX40L[59]. The innate response therefore sets the stage for the adaptive arm, in which dendritic cells present environmental and microbial antigens or autoantigens to naive T cells, promoting differentiation towards T_H2, T_H22 and, within certain endotypes, T_H17 lineages[60]. Notably, upstream alarmin blockade has been less successful clinically to date. Neither TSLP nor IL-33–ST2 inhibition has shown robust efficacy in phase II randomized trials[61,62].

IL-13 has emerged as an important cytokine in this immunological process, suppressing barrier gene expression, sensitizing pruriceptors via neuronal JAK1 and inhibiting antimicrobial peptide production[13,63–65]. This is validated by the clinical efficacy of selective IL-13 inhibitors such as tralokinumab and lebrikizumab[66,67] and IL-4Rα inhibition through dupilumab[68]. Complementing IL-4–IL-13, the IL-31–IL-31RA axis directly drives non-histaminergic itch and neuronal sensitization; clinical IL-31RA blockade (nemolizumab) confirms this pathway's relevance in AD[69]. Although type 2 circuits dominate, additional immune modules contribute to disease heterogeneity. IL-22 promotes impaired differentiation and epidermal thickening[14,63,70] and may mark subgroups that benefit from IL-22 blockade[71], while the relative contribution of T_H1 and T_H17 cell responses may vary across individuals and age groups and is potentially more relevant in chronic disease stages and subsets of people such as in Asian individuals, in whom AD has been associated with stronger T_H17–T_H22 cell skewing and more psoriasiform features[60,72]. However, IL-17A–IL-23 pathway blockade has not shown meaningful efficacy in AD[73–75]. The OX40–OX40L costimulatory signal promotes the proliferation, activity and survival of pathogenic effector and memory T cells[76], and clinical trials of OX40–OX40L inhibitors showed clinically meaningful responses with high durability in some patients, further supporting T cell survival signalling as a disease maintenance mechanism[77,78]. B cells further contribute by producing autoreactive IgE and IgG against keratinocyte antigens such as periplakin and α-NAC[79,80]. Nevertheless, IgE targeting has produced mixed or negative results in controlled trials in adults with AD, suggesting that autoreactive antibodies are either modulatory players or epiphenomena, rather than primary drivers of disease[81,82]. Emerging data also implicate IL-9 (notably in paediatric AD) and IL-18 pathways in disease amplification; preliminary data indicate that IL-9 may increase IL-18 responsiveness in T_H2 cells, thereby amplifying type 2 inflammation in lesional skin[83].

A major conceptual advance is the recognition that tissue-resident memory T (T_{RM}) cells serve as a pivotal adaptive reservoir linking past inflammation to future relapses. T_{RM} cells ($CD4^+$ and $CD8^+$) persist in clinically 'resolved' skin within epidermis–dermis niches, retaining a poised effector phenotype capable of rapid cytokine production consistent with T_H2 cell, and also T_H22 and T_H17 cell, signalling[84]. Upon barrier injury, allergen exposure or microbial insult, signals from both the innate compartment (for example, group 2 innate lymphoid cell-derived cytokines) and adaptive pathways converge to awaken T_{RM} cells, precipitating site-specific flares. Concomitantly, keratinocytes in previously

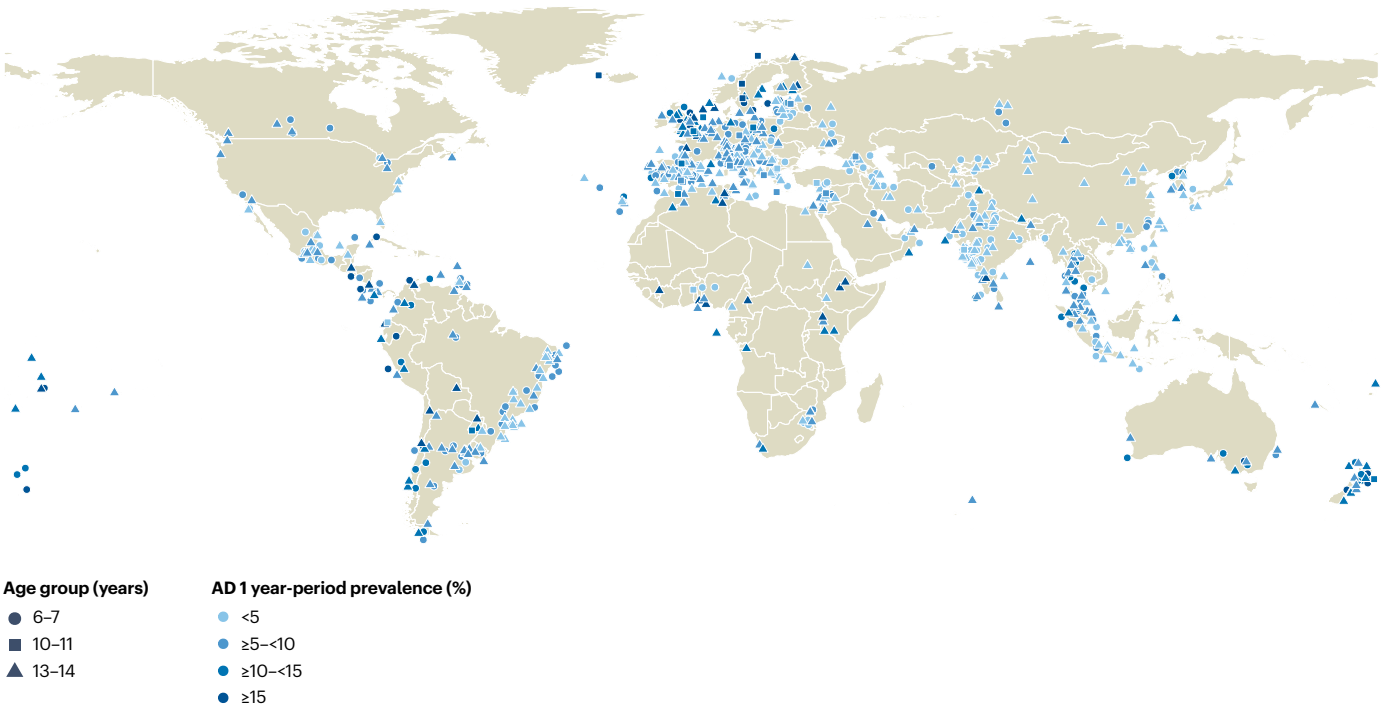


Fig. 2 | Atopic dermatitis prevalence in children. There is wide variation in the prevalence of atopic dermatitis (AD) across the world, with particularly high burdens seen in Northern Europe and other affluent, as well as urban, settings. Differences in prevalence levels between sites within the same country, such as in Spain and the UK, additionally point towards lifestyle and environmental (rather than genetic) risk factors. Points are randomly shifted along the longitude axis to reduce overlap and enhance visualization. Data from ref. 33. Map designed by and courtesy of S. Keddie, Global Atopic Dermatitis Atlas Coordinating Centre at King's College London, 2025.

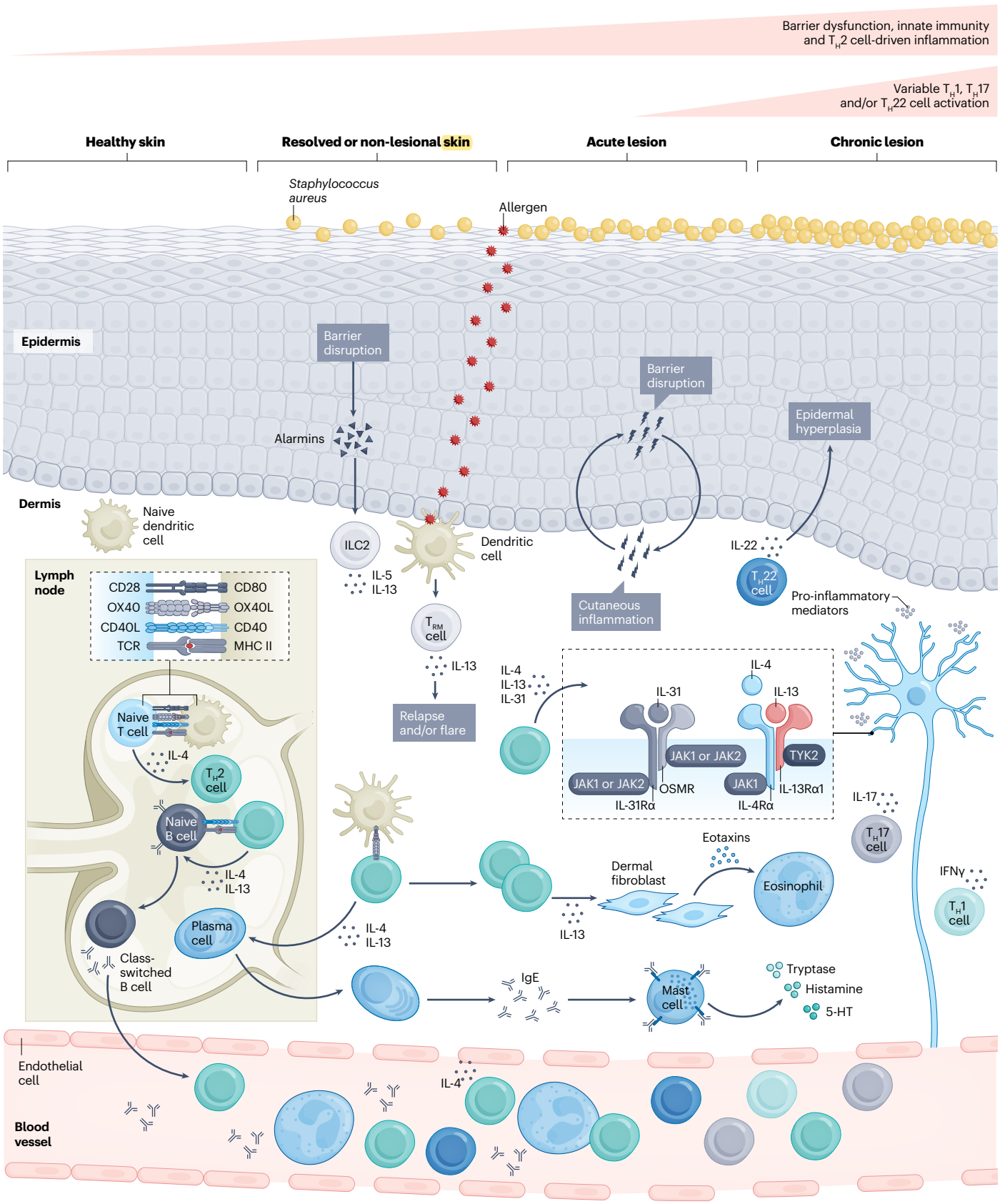

Barrier dysfunction, innate immunity and T_H2 cell-driven inflammation
Variable T_H1, T_H17 and/or T_H22 cell activation
Healthy skin
Resolved or non-lesional skin
Acute lesion
Chronic lesion
Staphylococcus aureus
Allergen
Epidermis
Barrier disruption
Barrier disruption
Epidermal hyperplasia
Alarmins
Dermis
Naive dendritic cell
ILC2
IL-5
IL-13
Dendritic cell
IL-22
T_H22 cell
Pro-inflammatory mediators
Cutaneous inflammation
T_{RM} cell
IL-13
Relapse and/or flare
Lymph node
CD28
CD80
OX40
OX40L
CD40L
CD40
TCR
MHC II
Naive T cell
IL-4
T_H2 cell
Naive B cell
IL-4
IL-13
Plasma cell
Class-switched B cell
IL-4
IL-13
IL-4
IL-13
IL-31
IL-31
IL-4
IL-13
JAK1 or JAK2
TYK2
JAK1 or JAK2
OSMR
JAK1
IL-13Rα1
IL-31Rα
IL-4Rα
IL-17
T_H17 cell
IFNγ
T_H1 cell
Eotaxins
Dermal fibroblast
Eosinophil
IL-13
IgE
Mast cell
Tryptase
Histamine
5-HT
Endothelial cell
IL-4
Blood vessel

Fig. 3 | Stage-based pathogenesis and key immune circuits in atopic dermatitis. The pathophysiological mechanisms underlying atopic dermatitis (AD) evolve across disease stages, beginning with subclinical barrier abnormalities and inflammation, and progressing to acute and chronic inflammatory lesions driven by interacting epithelial, immune, microbial and neuroimmune pathways. Even clinically unaffected AD skin shows subtle barrier impairment and microbial dysbiosis, including increased *Staphylococcus aureus* colonization, with keratinocyte-derived alarmins (TSLP, IL-33 and IL-25) activating type 2 innate lymphoid cells (ILC2) and tissue-resident memory T (T_{RM}) cells to produce pro-inflammatory cytokines, such as IL-5 and IL-13. Allergens and microbial antigens that penetrate the epidermis are captured by Langerhans cells, inflammatory epidermal dendritic cells and dermal dendritic cells. These cells engage co-stimulatory signals (CD28–CD80, OX40–OX40L and CD40L–CD40) and prime naive T and B cells in lymph nodes, leading to T helper 2 (T_H2) cell polarization, class-switch recombination and IgE production by plasma cells. IgE subsequently binds to FcεRI on mast cells, thereby sensitizing them, and allergen-mediated crosslinking releases mediators that can amplify local inflammation and itch. In acute lesions, T_H2–T_H22 cell cytokines (notably IL-4, IL-13 and IL-22) further disrupt barrier function, drive inflammation and pruritus, and – together with fibroblast-derived eotaxins – recruit eosinophils. Signalling through IL-4–IL-13 and IL-31 receptors engages JAK–STAT pathways across immune, epithelial and neural cells, reinforcing neuroimmune itch circuits. In persistent disease, chronic lesions exhibit epidermal hyperplasia and dermal remodelling with a mixed T cell infiltrate and variable contributions of T_H1 and T_H17 cell responses alongside dominant type 2 responses. 5-HT, 5-hydroxytryptamine.

inflamed sites retain stress-induced programmes with reduced barrier gene expression (for example, filaggrin). This interplay between innate sentinel responses, adaptive immune memory and persistent epithelial reprogramming forms the mechanistic basis for the chronicity and relapse pattern of AD and is complemented by emerging evidence for 'inflammatory memory' programmes in keratinocytes that render barrier genes more susceptible to rapid dysregulation upon re-challenge[85].

Neuro-immune–barrier axis of itch and inflammation

Histamine-independent pathways are central to chronic itch in AD, with cytokine–JAK–STAT signalling an essential pathway in sensory neurons. Neuronal JAK1 acts as a key mediator of type 2 cytokine-driven itch: IL-4 and IL-13 sensitize TRPV1- and TRPA1-positive pruriceptors and lower their activation thresholds to diverse pruritogens[65], whereas IL-31 directly activates IL-31RA–OSMR-expressing, TRPV1- and TRPA1-positive neurons to evoke itch[21] (Fig. 4). All three cytokines were reported to promote dorsal root ganglion sensory neuron growth and epidermal nerve growth; compared with IL-31, IL-4–IL-13 (via IL-4Rα) exerts broader neuroimmune and barrier effects[22]. Basophils and sensory neurons form reciprocal circuits, via leukotrienes and IL-4–IL-13, that amplify neuronal excitability[86]. Keratinocyte-derived artemin, regulated by the aryl hydrocarbon receptor, further contributes to epidermal hyperinnervation[87]. Therapeutically, blockade of IL-4Rα rapidly alleviates itch in AD, in part by silencing neuronal JAK1 signalling before full resolution of skin inflammation[68], while IL-31RA inhibition with nemolizumab directly disrupts the IL-31–neuron circuit, leading to reduced nocturnal scratching and improved sleep[88].

Skin microbiome, virome and mycobiome

AD is characterized by a profound dysbiosis that is broadly characterized by an increased bacterial load, a reduction in microbial diversity, and a higher presence and relative expansion of *S. aureus*[89]. Increased abundance of *S. aureus* is associated with higher severity and chronicity and is seen during flares[15,18,90]. Before AD onset, infants are not typically colonized with *S. aureus*; however, infants who later develop AD show a delayed maturation with slower diversification of the early skin microbiome than infants who do not develop AD[91,92]. *S. aureus* damages keratinocytes and tight junctions and triggers epithelial alarmins via cytolysins and proteases, and broadly activates T cells through superantigen exotoxins[19]. Furthermore, *S. aureus* can promote itch and persistence through pruritogenic toxins and virulence factors whose expression is coordinated by bacterial quorum sensing (that is, cell density-dependent regulation of toxin production)[16,19,20]. *S. aureus* is also a common source of secondary bacterial superinfection. Although there is no single AD-specific *S. aureus* clone, particular lineages, for example CC1 in several cohorts, and virulence gene sets including superantigen genes such as *sea*, *sec*, *sed* and *tsst-1*, seem to be enriched in AD[89]. It is commonly assumed that the epidermal barrier deficiency and type 2 cytokine-mediated suppression[17] of antimicrobial peptide expression in AD create a microenvironment that increases susceptibility to *S. aureus* colonization, which then in turn amplifies type 2-dominated inflammation and itch in a bidirectional loop[93]. This is supported by observations that effective treatment, in particular via inhibition of type 2 signalling, restores dysbiosis and reduces cutaneous infections[94–97]. Treatments directly targeting *S. aureus*, either through attenuation of disease-driving clones or general promotion of more diverse microbiomes, are technically complex to develop and are still far from being available in the clinic.

In contrast to bacteria, fungal diversity has been reported to be higher in people with AD[17]. Cutaneous yeasts such as *Malassezia* species, which typically colonize sebaceous areas, can trigger or exacerbate inflammation in AD[98]. *Malassezia* species can promote skin inflammation by secreting proteases and inducing pro-inflammatory cytokine release from keratinocytes[99]. Moreover, specific IgE responses to *Malassezia* antigens, including antigens that crossreact with human homologues, are frequently detected in people with severe head-and-neck dermatitis[100,101]. Observational data and small randomized trials suggest that head-and-neck dermatitis may benefit from oral azoles in sensitized patients, while topical azoles show variable efficacy[102]. Exacerbations or de novo development of head, neck and/or facial involvement has been reported in a subset of patients with AD treated with dupilumab[103] and has been linked to increased baseline *Malassezia*-specific IgE[104].

Individuals with AD are more susceptible to viral skin infections (for example, HSV infection, which may also cause eczema herpeticum, and molluscum contagiosum virus infection)[47]. This vulnerability is linked to the epidermal barrier perturbation together with the complex inflammatory milieu; upregulation of the type 2 cytokine, interferon and IL-36γ inflammatory gene pathways alongside dysregulation of epidermal differentiation complex (barrier) genes have been reported in non-lesional skin of people with AD with a history of eczema herpeticum[105]. Consistent with a functional role for type 2 signalling in blunting antiviral defence, IL-4–IL-13 blockade with dupilumab reduces HSV type 1-specific IgE and enhances HSV type 1-specific type 1 cytotoxic responses in AD[106].

Diagnosis, screening and prevention

Diagnosis

Diagnostic criteria. Multiple diagnostic criteria have been developed for use in clinical practice. The Hanifin and Rajka, the UK Working Party

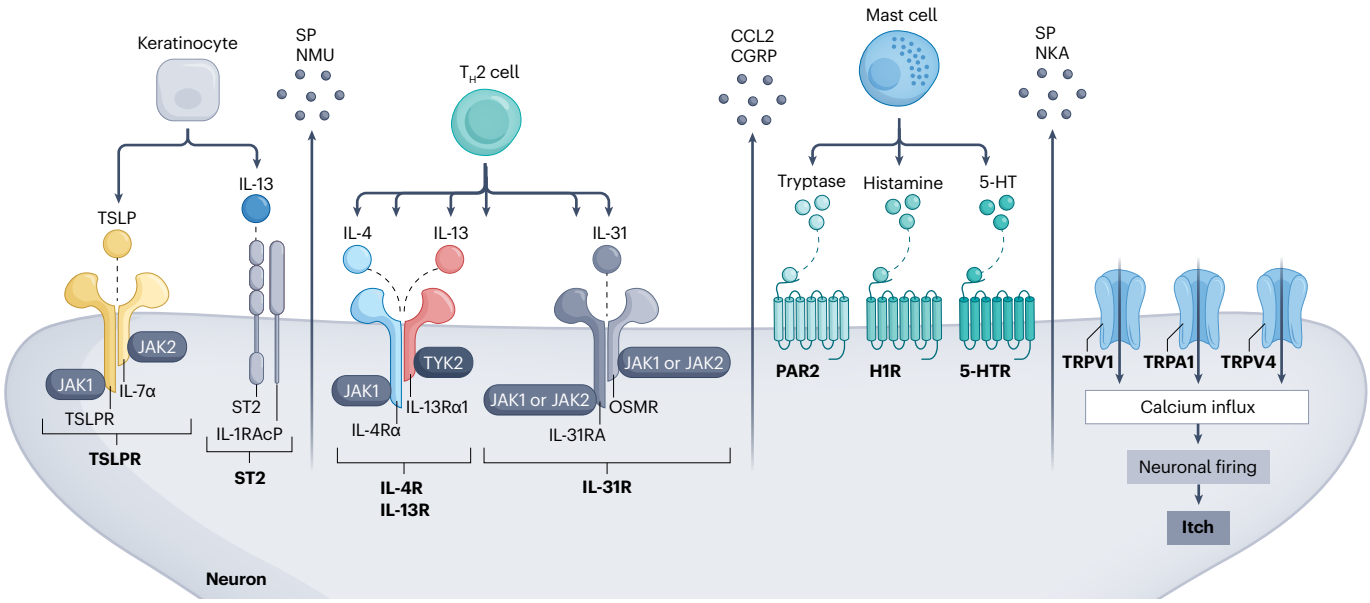


Fig. 4 | Neuro-immune pathways driving itch in atopic dermatitis. Itch in atopic dermatitis (AD) is driven mainly by non-histaminergic pathways, in particular IL-31, IL-4–IL-13, TSLP and IL-33, whereas histamine H1R has a minor, context-dependent role. Keratinocyte-derived alarmins (TSLP and IL-33) activate and/or sensitize pruriceptive neurons via TSLPR (JAK1 and JAK2) and ST2–IL-1RAcP and promote type 2 inflammation. Type 2 innate lymphoid cells release IL-5 and IL-13, and T helper 2 (T_H2) cells provide IL-4, IL-13 and IL-31. IL-4–IL-13 signal through the shared type 2 receptor (IL-4Rα–IL-13Rα1, JAK1–TYK2) to sensitize neurons and heighten excitability, whereas IL-31 directly evokes itch via IL-31R (IL-31RA–OSMR, JAK1 or JAK2). The mast cell mediators tryptase, histamine and 5-hydroxytryptamine (5-HT) act via PAR2, H1R and 5-HTR. Sensory neurons release chemokines and neuropeptides, such as CCL2, CGRP, substance P (SP), neurokinin A (NKA) and neuromedin U (NMU), which further amplify mast cell–neuron communication and broader neuroimmune crosstalk, including feedback into type 2 circuits such as type 2 innate lymphoid cell activation. Despite a reduction in intraepidermal nerve fibre density, AD skin shows increased neurite sprouting, upregulation of itch receptors and lower activation thresholds. Convergent inputs open transient receptor potential channels (TRPV1, TRPA1 and TRPV4), causing Ca^{2+} influx, action potential firing and itch.

refinement of the Hanifin–Rajka criteria, and the American Academy of Dermatology consensus criteria are the most frequently utilized, each defining AD using different combinations of signs, symptoms and patient history. Despite their differences, these criteria generally emphasize three diagnostic features: pruritus, a family history of atopic disease, and age-appropriate distribution pattern — typically flexural involvement[29,107,108]. Pruritus is widely regarded as the hallmark feature of AD and is mandatory in both the UK Working Party and American Academy of Dermatology definitions, as well as being one of the four major criteria in the Hanifin and Rajka definition, in addition to a typical age-related morphology and distribution of lesions, a chronic or chronically relapsing disease course, and a personal or family history of atopy[107]. In addition to the core diagnostic features, several minor or non-essential criteria are included, such as early age at onset, xerosis (dry skin) and specific IgE responses[7]. Features such as lichenified dermatitis, nipple dermatitis and prurigo nodularis-like lesions are inconsistently reported in patients, and may reflect phenotypic variants[8].

Existing criteria have been criticized for limited applicability across age groups and skin tones[109,110]. For example, pruritus can be difficult to assess in infants and is reportedly less prominent in older adults[2,111]. In people with richly pigmented skin, conventional clinical signs such as erythema, may be less apparent, whereas papulation, lichenification, follicular accentuation and postinflammatory dyspigmentation may be more prominent, creating diagnostic challenges and potentially leading to underestimation of disease severity[112]. Because no alternative diagnostic criteria have yet been universally validated for darker skin, these limitations are best addressed by careful attention to morphology and distribution, palpation, patient-reported itch and improved training in the recognition of inflammatory skin disease across diverse skin tones.

Clinical phenotype. The clinical phenotype of AD is highly variable and influenced by age, ethnicity and skin tone (Fig. 5). Age-dependent distribution patterns have been described. In infants, lesions often present on the cheeks, neck, scalp and extensor surfaces, not just flexurally[1]. In children (2–12 years old), adolescents (12–18 years old) and young adults (18–39 years old), AD is typically found on the flexural areas, with neck involvement also being common[1,2]. By contrast, periorbital involvement and lesions on the anterior neck and hands are more common in older adults (≥40 years old)[2].

Ethnic and regional differences also contribute to phenotypic heterogeneity. In individuals of African descent, AD can present with papules and perifollicular localization on the trunk and extensor surfaces[113]. Prurigo nodularis and postinflammatory hyperpigmentation are more common in individuals with darker skin tones than in those with lighter skin tones[114]. In Asian populations, such as in India, nummular and follicular AD phenotypes typically predominate[113]. These clinical differences may be linked to microbiome signatures, and are possibly also related to distinct immune responses, including increased T_H17–T_H22 cell activity in Asian populations and predominant T_H2–T_H22 cell responses in people of African descent[72,115]. A better definition of such phenotypic and biological heterogeneity should support more precise clinical

trial design, including studies that specifically address well-defined subtypes, such as hand-and-foot AD and head-and-neck AD.

Severity scoring. Multiple severity scoring systems exist for AD: the most used include the EASI and SCORAD criteria[116,117]. Both assess lesional severity based on the presence of clinical features such as erythema, oedema, lichenification and excoriation, with the additional inclusion of oozing–crusting and dryness quantifiers within the SCORAD criteria[118,119]. A key difference between the two systems is the addition of a patient's symptoms (itch and sleep loss) in SCORAD[119]. Patient-oriented severity scoring systems are also used and include the Patient-Oriented Eczema Measure and the Patient-Oriented SCORAD scores[120,121]. The latest version of the Patient-Oriented SCORAD has been specifically designed for patients with different skin colours to address its former inadequacy in assessing AD severity in those with richly pigmented skin types[119].

There is a growing case for using artificial intelligence (AI) in the diagnosis and severity scoring of AD. Studies have shown the high diagnostic accuracy of AI combined with sophisticated image analysis techniques in distinguishing AD from its differential diagnoses[122]. Integrating AI into diagnostic workflows may help reduce the subjectivity and bias inherent in clinician assessments; however, it remains essential to ensure that AI systems are trained adequately on darker skin phototypes to avoid perpetuating diagnostic disparities[123].

Prevention

Primary prevention. Various primary prevention measures have been considered in parallel with advances in the understanding of the immunopathology of AD. One particular area of scientific debate is the hypothesis that there is a lower risk of AD in populations exposed to higher levels of certain pathogens, including helminth parasites, and with direct contact with farm animals, especially during the perinatal period ('biodiversity hypothesis')[124–126]. Similarly, a lower risk of AD and allergic sensitization has been observed among people who consumed unpasteurized milk in childhood[125,127]. This may also partly explain why the burden of AD is greatest in urbanized, high-income regions[5].

Another important protective factor is a diverse gut bacterial microbiome, which is reduced in those born via caesarean section or after antibiotic exposure during the last trimester of pregnancy and in early life[128–131]. In addition, breastmilk intake is associated with a dose-dependent increase on the bacterial diversity of an infant's gut microbiome, and the promotion of breastfeeding has a protective effect on the development of AD even into adolescence[132,133]. However, there is no evidence that prolonging exclusive breastfeeding without oral food intake beyond 3–4 months of age has a protective effect. The emerging data linking AD and the gut microbiome could point towards a prevention strategy. For example, the potential use of probiotic supplementation during pregnancy and infancy to reduce AD prevalence, although interpretation and reproducibility of such studies are hampered by methodological diversity[134,135].

Infants who subsequently develop AD have distinct skin barrier and microbiome changes compared with infants who do not develop AD[136]. This knowledge has led researchers to test primary prevention strategies targeting the skin barrier, for instance through regular application of moisturizers in early life. While this approach has been unsuccessful with standard emollients, more specialized preparations may still have a beneficial effect[137–139]. Water hardness (high calcium



Fig. 5 | Diverse clinical presentations of atopic dermatitis. Atopic dermatitis has a broad range of clinical presentations. **a**, Nodular changes. **b**, Facial eczema in an infant with marked xerosis. **c**, Discoid (nummular) eczema. **d**, Extensor involvement. **e**, Severe lichenification. **f**, Punched out lesions in eczema herpeticum with secondary impetiginization. **g**, Facial eczema in an older child. **h**, Follicular accentuation. **i**, Excoriations, healing herpes simplex virus infection and leukoderma. **j**, Chronic lichenification with postinflammatory hyperpigmentation and leukoderma. Panels **b**, **e** and **g**, images courtesy of GADA/ILDS. Panel **h** reprinted from ref. 297, CC BY 4.0.

carbonate content in domestic water) has also been shown to increase the risk of AD development, especially in those carrying a *FLG* LoF mutation[140,141]. This may, in part, be due to a higher deposition of surfactants from wash products on the surface of the skin after washing. Thus, the installation of a water softener before birth in areas of hard domestic water could help to prevent AD development, but a multicentre definitive trial is required to establish this as a valid prevention method[142].

Prevention of atopic multimorbidity either through either preventing AD ab initio or by treating AD early in the disease course to prevent accumulation of these comorbidities, including asthma, allergic rhinitis and food allergy, would represent a substantial public health advance. Primary prevention is the subject of many ongoing initiatives but to date no convincing evidence of its effectiveness has been provided.

Secondary prevention. Practical guidelines recommend secondary prevention strategies based on trigger avoidance[143]. Key triggers for AD flares include environmental irritants, food allergens, aeroallergens and sweat, with avoidance strategies dependent on disease stage and the individual patient[144–146]. The spectrum of relevant allergens changes with disease trajectory. Environmental irritants, such as air pollutants, tobacco smoke and surfactants contained in wash products, such as sodium lauryl sulfate, can affect individuals with sensitive skin and lead to AD exacerbations[144,147]. Food allergens can be particularly important in infants with early-onset, more severe AD, and should be investigated further if there is a supporting history of immediate-type reactions[148]. Caution should especially be expressed when eliminating food from diets of children with AD without a clear history of immediate-type reactions, as early dietary introduction has been shown to induce tolerance[149,150].

By contrast, environmental allergens, such as house dust mites, pollen or animal fur, seem to be more relevant as trigger factors in older children and adults[151]. Avoidance recommendations, such as encasing mattresses and pillows, should be suggested for individuals with proven allergic sensitization[152]. Even in these individuals, the effects of avoidance strategies may still be marginal.

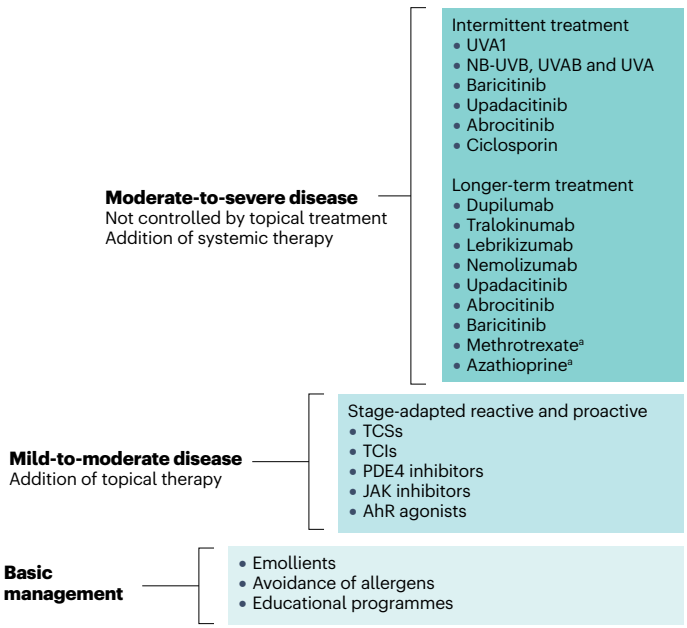


Fig. 6 | Key steps in atopic dermatitis management. According to current guidelines and the expert clinical opinion of the authors, the management algorithm reflects a cumulative and flexible approach. In this approach, treatment components may be added, intensified or tapered according to disease activity and individual patient needs, rather than replaced in a strict stepwise manner. The algorithm is intended to assist, not replace, clinical decision-making. AhR, aryl hydrocarbon receptor; NB-UVB, narrowband Ultraviolet B; PDE4, phosphodiesterase 4; TCIs, topical calcineurin inhibitors; TCSs, topical corticosteroids; UVA, Ultraviolet A; UVAB, Ultraviolet A and Ultraviolet B; UVA1, Ultraviolet A1. [a]Not approved for the treatment of atopic dermatitis; current guidelines provide a conditional recommendation for their use.

Management

The immediate goals of AD management are to reduce symptoms, particularly itch, and improve skin signs, whereas longer-term goals are sustained disease control that enables patients to function fully at home, at work and at school, and, in some patients, remission (Box 1). As outlined in current guidelines across several countries and professional societies[153–159], core elements are patient education, avoidance of trigger factors, restoration and maintenance of the skin barrier, and anti-inflammatory topical therapy matched to age, site and severity, with escalation to phototherapy or systemic agents when disease cannot be adequately controlled with topical treatment alone (Fig. 6).

Over the past decade, the therapeutic landscape has changed markedly, with non-steroidal topicals, injectable biologics and oral JAK inhibitors (JAKi) expanding options across ages and severities so that escalation is no longer limited to broad immunosuppression. In parallel, treatment goals have become more ambitious; beyond control and flare prevention, sustained remission is increasingly realistic for many patients, and early, mechanism-based intervention raises the prospect, still under study, of modifying the disease course (Box 1). However, the practical impact of these advances differs widely across health systems. In many low-income and middle-income countries, access to advanced therapies remains limited (often shaped by reimbursement pathways and the availability of monitoring infrastructure), and care therefore still relies heavily on optimized topical anti-inflammatory therapy, patient education, phototherapy where available, and conventional systemic agents[160,161].

Basic management

Generalized xerosis and irritable skin are present in most people with AD, and reflect inherited and acquired defects in epidermal structure and function, associated with increased transepidermal water loss, reduced hydration, pH shifts and increased permeability. Regular, liberal use of moisturizers is widely recommended despite limited trial evidence, as it improves hydration and xerosis, reduces itch, prolongs remission and lowers the need for anti-inflammatory treatment[162]. Emollients combine occlusive agents (for example, petrolatum and liquid paraffin) that reduce water loss with humectants (for example, glycerol, urea and lactic acid) that increase water binding. Barrier lipid-enriched formulations containing ceramides or essential fatty acids are biologically plausible, but comparative trials have not consistently demonstrated clinically meaningful superiority over standard emollients, and no specific formulation can currently be recommended routinely[163]. In practice, bland emollients with few ingredients and without fragrances or perfumes, selected according to skin type, body site, degree of dryness and inflammation, as well as patient preference and cost, are advised[164,165].

Evidence on hygiene measures is limited, and guidelines provide few firm recommendations beyond avoiding soaps and conventional shower gels. There is no demonstrated benefit of emollient bath additives[166] and it has been reported that frequent bathing or showering does not worsen AD severity[167]; a crossover trial in children showed that twice-daily bathing followed by emollient application outperformed twice-weekly bathing without emollient, suggesting that consistent moisturizer use after bathing is more important than bathing frequency[168].

Given the strong association between *S. aureus*-dominant dysbiosis and disease activity, adjunctive measures to reduce microbial burden (for example, antiseptic approaches) can be considered in selected patients (for example, those with frequent bacterial superinfections), whereas systemic or topical antibiotics should generally be reserved for clinically overt infection to avoid collateral microbiome disruption. Emerging microbiome-sparing strategies (including topical prebiotics and postbiotics, live biotherapeutic products and engineered commensals) are under clinical evaluation but are not yet part of routine care[169].

Topical therapy

Conventional topical anti-inflammatory therapy. Topical anti-inflammatory therapy is a core element of AD management and is sufficient to control mild-to-moderate disease in most patients. Topical corticosteroids (TCSs) remain the first-line option across all ages to treat acute exacerbations, and their efficacy is well established in randomized controlled trials (RCTs) and network meta-analyses[170,171]. The choice of potency, formulation and vehicle, and the treatment schedule, are guided by disease activity, patient age, body site and planned duration. Local adverse effects include atrophy (skin thinning), striae (stretch marks), telangiectasias (visible small dilated blood vessels), purpura (purple discolouration caused by small skin bleeds), dyspigmentation (skin colour changes) and acneiform changes, particularly with prolonged use of high-potency preparations[170]. Systemic effects are uncommon and relate to extended use of high-potency agents over large areas[172]. Patient concerns and misconceptions about TCSs are common, and may affect adherence and real-world outcomes[173]; adverse events are largely avoidable with appropriate use[170,171]. For flares, application once or twice daily for up to 2–4 weeks is typically recommended, followed by reduction in potency or frequency. Proactive intermittent application (for example, twice weekly to previously involved sites) reduces relapses and prolongs remission[174]. Lower potencies are preferred on areas of sensitive skin such as the face, and intertriginous and genital areas, and in children, except for short, supervised courses during severe flares.

Topical steroid withdrawal syndrome (also referred to as red skin syndrome) has been described as a severe rebound reaction after prolonged use of more potent TCSs, characterized by extensive redness, pain and skin peeling, often with a sharp cut-off at wrists and ankles[175]. Its aetiology and exact diagnostic criteria remain unclear, and robust epidemiological data are lacking. Increasing awareness of topical steroid withdrawal syndrome may aid early recognition and prevention, but can also fuel steroid hesitancy; further research is needed to better define this entity and its optimal management.

The topical calcineurin inhibitors tacrolimus 0.03% or 0.1% ointment and pimecrolimus 1% cream are safe, steroid-sparing treatments suitable both for proactive maintenance after a short induction with a topical anti-inflammatory (usually a TCS) and for first-line use on sensitive areas such as the face and intertriginous or genital skin. The anti-inflammatory activity of tacrolimus 0.1% is broadly comparable to that of a moderately potent TCS, whereas the activity of pimecrolimus is closer to that of a mildly potent TCS[176]. In routine care, many clinicians alternate TCSs and topical calcineurin inhibitors by site and treatment phase to balance efficacy, tolerability and patient preference. Topical calcineurin inhibitors do not cause skin atrophy; the most common adverse effect is transient application site stinging[176].

Novel anti-inflammatory therapies. Newer non-steroidal topical agents, including phosphodiesterase 4 (PDE4) and JAKi, provide steroid-sparing options with distinct tolerability and safety profiles, and can be particularly useful when long-term treatment is anticipated, local adverse effects of TCS are a concern, or acceptance of topical steroids is low. However, their use is currently constrained by regional differences in approval, availability and cost.

Topical PDE4 inhibitors (for example, crisaborole and roflumilast) are approved in some countries for mild-to-moderate AD, and show efficacy broadly comparable to that of mild TCS, with pruritus reduction and good safety; application site burning or stinging is the main adverse effect with crisborole and is significantly less often reported with rofumilast[170,177–180]. Where affordable and accessible, their differentiation will be attractive for patients and parents hesitant to use topical steroids.

Topical JAKi such as ruxolitinib cream and delgocitinib ointment provide potent local anti-inflammatory effects with rapid itch relief and efficacy comparable to those of moderate-potency TCS, with favourable local tolerability and low systemic exposure when used according to the label[181–186].

Tapinarof cream 1% is a non-steroidal aryl hydrocarbon receptor agonist, approved in the USA for once-daily treatment of AD, combining anti-inflammatory effects with induction of epidermal differentiation gene expression via aryl hydrocarbon receptor–NRF2 signalling. Phase III trials showed clinically meaningful EASI and Investigator Global Assessment responses in about half of treated patients, with folliculitis and mild application site reactions as the main adverse events[187,188].

Systemic therapy

Phototherapy and conventional systemic treatments. If disease control cannot be achieved with topical measures, ultraviolet light therapy or systemic therapy is needed. In a 2021 Cochrane review[189], low-certainty evidence suggested that narrowband Ultraviolet B, compared with placebo or no treatment, improved clinician-rated signs after 12 weeks (mean difference −9.4 on a 0–90 disease activity scale) and increased the likelihood of itch improvement (risk ratio 1.72, 95% CI 1.10–2.69). By contrast, evidence for Ultraviolet A1 or PUVA (psoralen and Ultraviolet A) was of very low certainty, with no reliable conclusions on comparative efficacy because the available studies were small and imprecise. Reported short-term harms were uncommon overall. Depending on cumulative doses, phototherapy carries the risk of photodamage and cutaneous malignancy[189]. Phototherapy is limited by accessibility and convenience because it usually requires frequent treatment visits, travel and substantial time commitment over prolonged periods.

Over decades, for lack of alternatives, conventional immunosuppressants — ciclosporin and methotrexate, and in selected refractory cases azathioprine or mycophenolate mofetil — were repurposed to treat particularly severe AD; they remain widely used and are still the first-line systemic option in many countries owing to their low cost

and more favourable reimbursement compared with newer targeted agents[190]. Overall, however, trial evidence for their long-term efficacy and safety is limited. Both ciclosporin and methotrexate seem effective but carry important safety risks and require close clinical and laboratory monitoring throughout treatment. In many countries only ciclosporin is licensed for short-term treatment of severe, therapy-refractory AD in those aged ≥16 years. Due to cumulative nephrotoxicity it should not be used for longer than 1–2 years at a stretch and ideally in short or intermittent courses[155–159]. Ciclosporin acts rapidly, with ~55% mean improvement in disease severity after 6–8 weeks[191]. Methotrexate has a slower onset of action, with clinical responses typically emerging after 8–12 weeks and about 40–50% of patients achieving good or excellent control in longer-term studies[192], and it can often be continued for years with appropriate monitoring. Accumulating evidence from observational cohorts and RCTs in paediatric patients suggests sustained disease control in some patients[193].

Advanced systemic therapy. With the growing number of approvals for targeted systemic therapies, conventional immunosuppressants are decreasing in popularity in higher income countries but accessibility to these treatments is limited in many countries and, even in many advanced economies, access depends on treatment failure or intolerance of conventional immunosuppressants. An analysis from the TREAT registry in Germany, where advanced therapies can be used first-line in eligible patients, showed a marked shift after the introduction of dupilumab and subsequent targeted agents. Until 2017, only around one-third of registry patients with moderate-to-severe AD had received systemic treatment and >80% of those had received broad immunosuppressants; however, by 2021, the proportion receiving systemic therapy had increased to more than two-thirds and >80% of systemically treated patients had received an advanced systemic therapy[194]. However, as noted above, in many countries, access to modern agents remains contingent on prior immunosuppressant use for cost reasons; therefore, systemic care still relies largely on conventional agents, with treatment choice often driven by cost, monitoring requirements and local infrastructure.

Biologics. Currently approved biologic treatments for AD target downstream type 2 effector cytokines. Dupilumab is a human IgG4 monoclonal antibody to IL-4Rα that inhibits signalling of both IL-4 through the type 1 receptor on haematopoetic and immune cells and the type 2 receptor on keratinocytes, sensory neurons and other cells[195]. Tralokinumab (fully human IgG4) and lebrikizumab (humanized IgG4) both target soluble IL-13 but bind different epitopes and with different affinities[195,196]. The humanized IgG2 antibody nemolizumab blocks IL-31 signalling by antagonizing IL-31RA, thereby dampening pruritogenic neuronal activation and downstream inflammation. It is also approved for the treatment of prurigo nodularis[197].

Dupilumab has been a breakthrough in the treatment of AD, and is approved for use in patients from 6 months of age. Numerous RCTs and extension studies, together with several network meta-analyses, have demonstrated robust efficacy with mean reductions in EASI of ~70% with 70% showing reductions of 50% (EASI-50) and 50% showing reductions of 75% (EASI-75) at week 16 on monotherapy, and clinically relevant itch improvement in ~40% of patients; responses increase with concomitant TCS and longer-term therapy[198–202]. Overall safety is favourable, with ocular surface complaints (particularly conjunctivitis and blepharitis) being the most frequent adverse events (up to 30–40% in trials and real-world cohorts); they are usually transient and manageable without drug withdrawal, but can be the reason for treatment discontinuation[199]. Routine laboratory monitoring is not required[203]. Efficacy and safety in children and adolescents are broadly comparable to those in adults[204,205], and real-world studies have shown high drug survival over several years with low discontinuation due to adverse events or insufficient efficacy[194,206–210]. Mechanistic studies during dupilumab therapy show broad normalization of inflammatory and barrier-associated signatures and improvements in microbial ecology, consistent with clinical benefit[25,70,94,96,211–214]. Beyond AD, dupilumab has approvals across several type 2 comorbidities (including asthma, chronic rhinosinusitis with nasal polyps, prurigo nodularis, chronic spontaneous urticaria and eosinophilic oesophagitis), with labels and age cut-offs varying by region.

Across pivotal trials and recent network meta-analyses, the short-term efficacy of lebrikizumab on skin signs and itch seems broadly comparable to that of dupilumab, whereas tralokinumab tends to achieve slightly lower responses[200,215,216]. Both IL-13 inhibitors are approved for use in patients from 12 years of age and have safety profiles similar to that of dupilumab, with conjunctivitis and injection-site reactions among the most frequent adverse events. Tralokinumab has also been associated with target-consistent transcriptomic and microbiome changes in lesional skin[217–219]. Nemolizumab provides rapid and clinically meaningful itch relief but more modest improvements in skin lesions; meta-analyses indicate a generally favourable safety profile versus placebo[200,220,221].

JAKi. Oral small-molecule JAKi modulate signalling downstream of multiple cytokine receptors implicated in AD and other immune-mediated diseases. Of the three agents currently approved for the treatment of moderate-to-severe AD, upadacitinib and abrocitinib preferentially inhibit JAK1, whereas baricitinib inhibits JAK1 and JAK2 to a comparable extent. In many countries, upadacitinib and baricitinib are approved for additional indications, which may guide treatment choice in patients with immune-mediated comorbidities.

RCTs, network meta-analyses and real-world data consistently demonstrate high short-term efficacy of upadacitinib, abrocitinib and baricitinib in AD[200,216,222,223]. In head-to-head trials, higher-dose upadacitinib and abrocitinib achieved faster and initially greater improvements in skin signs and itch than dupilumab[224,225]. Network meta-analyses generally rank upadacitinib and higher-dose abrocitinib among the most efficacious systemic options on trial end points (typically 12–16 weeks)[198,200,202]; longer-term effectiveness is better captured by extension studies and real-world persistence data, which suggest that between-class differences may attenuate with increasing treatment duration[201,209,226–228]. Tolerability and safety of JAKi in AD are overall favourable but more complex than with biologics. Across trials, meta-analyses and population-based cohorts, JAKi are associated with small but consistent increases in the risk of herpes zoster, acne or acneiform eruptions, and laboratory abnormalities (including dyslipidaemia and elevations in liver enzymes and creatine kinase)[229,230].

All systemic JAKi carry class warnings for venous thromboembolism, major adverse cardiovascular events, serious infections and malignancy, largely extrapolated from tofacitinib data in rheumatoid arthritis[231]; to date, clear signals for these outcomes have not emerged in RCTs in patients with AD or in meta-analyses[229,232,233]. Long-term use therefore requires individualized benefit–risk assessment, appropriate baseline screening and ongoing laboratory monitoring, particularly in older patients and those with cardiovascular risk factors.

Given their rapid onset and short half-lives, short-term to mid-term or intermittent use, flexible dosing and maintaining the lowest effective maintenance dose are being explored as strategies to reduce cumulative exposure, although prospective evidence is still limited and current recommendations are largely based on expert guidance[158,234,235].

Difficult-to-treat areas

The involvement of certain anatomical locations (for example, head, neck or face and hands or feet) can be particularly burdensome, and may be challenging to manage with topical therapy alone, even when global disease scores or body surface area affected appear only moderate[236]. In clinical practice and across treatment frameworks, involvement of these sites is often considered when escalating to advanced systemic therapy. Dedicated trials addressing such locations are emerging; for example, dupilumab showed substantial benefit in moderate-to-severe atopic hand and/or foot dermatitis in the phase III LIBERTY-AD-HAFT study[237]. Location-focused studies for IL-13-targeted therapies in hand-and-foot-predominant AD have been completed for tralokinumab in atopic hand eczema (NCT05958407)[238] and are ongoing for lebrikizumab in atopic hand and foot dermatitis (NCT06921759)[239].

Biomarker-guided treatment choice

The choice of advanced systemic therapy is currently guided mainly by clinical factors, such as age, comorbidities, desired speed of response, pregnancy considerations, route preference, and safety and monitoring constraints[240]. In routine practice, a minority of patients exhibit partial responses or loss of response over time and require dose adjustment or switching between classes, consistent with mechanistic heterogeneity beyond a purely type 2-dominant model. A 487-gene expression profile in lesional skin scrapings identified a subgroup with a transcriptomic 'JAKi responder' pattern and comparatively weaker responses to type 2-targeted therapy[50]. Other blood-based and skin-based transcriptomic studies have likewise supported molecularly distinct subgroups with differences in response distributions, including signatures associated with poor outcomes under dupilumab[24,25].

Quality of life

People with AD often report a profound impact on quality of life, and psychological comorbidities, such as depression, anxiety and social withdrawal, are common. This is in part driven by sleep disturbance from the itch–scratch cycle, leading to reduced productivity at school or work, and the economic impact on individuals and their families. Sleep deprivation and a disrupted circadian rhythm also lead to sympathetic nervous system overactivation, which drives the cutaneous inflammation further[3,5,241]. People with AD have higher incidences of anxiety, depression, suicidal ideation, attention deficit hyperactivity, and autism spectrum disorder[34,242]; therefore, increasing the psychosocial impact on individuals and families. Psychosocial management, including cognitive behavioural therapy and group education sessions, of AD is an important component of holistic care and is the subject of a recent Cochrane review[243]. Quality of life (QoL) measurements are relevant for monitoring disease progression and evaluating the efficacy of interventions; they should include not only the current assessment but also the long-term outcomes of the disease.

QoL should be assessed according to the patient's age. QoL measures for skin diseases have been used in clinical trials, as demonstrated by systematic reviews[27,30], with emphasis on the Dermatology Life Quality Index[244], described in 1994 for adults, the Children's Dermatology Life Quality Index[245], created in 1995 for the paediatric population, the Infants' Dermatitis Quality of Life Index for infants, the Dermatitis Family Index and the Family Dermatology Life Quality Index[27,30]. Such instruments have been used across different geographic regions and cultural settings, although cross-context interpretation requires caution because QoL constructs are influenced by cultural, societal and health system factors.

Infants with severe AD need more time in day care to facilitate demanding skin care requirements, and their parents report more missed workdays[246]. Children with AD show reduced QoL, similar to children with asthma and diabetes mellitus[37]. Complaints such as itch, soreness, sleep disturbances with impairment of intellectual functioning, mood swings, limitation to physical activities such as swimming, and family social dysfunction are frequent. Two out of three children (5–17 years of age) with AD show sleep disturbance, which is ~2.3 times more common in children with moderate to severe disease than in those with mild disease[39]. A cross-sectional study in 2,553 children with AD in the USA using Columbia Impairment Scale scores found an association between AD and behavioural and functional problems; in this study, higher Columbia Impairment Scale scores were notably linked to male sex, older age, lower household income, public insurance, and comorbid depression and anxiety[40]. Teenagers with AD report a negative impact of the disease on relationships, self-image and sexual intimacy, and nearly 40% of school-aged children and teens with AD have experienced appearance-based bullying or stigmatization[247]. In adults with AD, a study in the USA showed that itch, dryness and inflamed skin were the most burdensome symptoms; moreover, patients with AD reported lifestyle limitations (51.3% of patients), impact on activities (43.3%) and social isolation (39.1%)[248]. In adults with AD, there was a correlation between poorer QoL and increased disease severity (SCORAD score)[11].

A systematic review and meta-analysis including 17 trials involving 6,665 patients with AD on dupilumab found significant substantial

Box 2 | HOME initiative-approved core outcome sets

The Harmonizing Outcome Measures for Eczema (HOME) initiative provides a core set of outcome measures to be reported in clinical trials[249]. It comprises four specific outcome domains, each of which has defined instruments for measurement.

Signs
- EASI (Eczema Area and Severity Index)

Symptoms
- POEM (Patient-Oriented Eczema Measure)
- Peak Itch NRS (numeric rating scale)

Quality of life
- DLQI (Dermatology Life Quality Index): ≥16 years
- CDLQI (Children's Dermatology Life Quality Index): 4–15 years
- IDQoL (Infants' Dermatitis Quality of Life Index): <4 years

Long-term control
- ADCT (Atopic Dermatitis Control Tool)
- RECAP (Recap of Atopic Dermatitis)

Table 1 | Therapeutic developments in atopic dermatitis

Drug	Mode of action	Development phase[a]	Clinical trial number (latest update)
Monoclonal antibodies			
Dupilumab	Anti-IL-4Rα	Regulatory approval	NA
Tralokinumab	Anti-IL-13	Regulatory approval	NA
Lebrikizumab	Anti-IL-13	Regulatory approval	NA
Nemolizumab	Anti-IL-31RA	Regulatory approval	NA
Stapokibart	Anti-IL-4Rα	Regulatory approval	NA
Amlitelimab	Anti-OX40L	III	NCT06130566 (2025)[265], NCT06181435 (2026)[266], NCT05769777 (2026)[267]
Bosakitug	Anti-TSLP	III	NCT05932654 (2025)[268]
Eblasakimab	Anti-IL-13Rα1	III	NCT05694884 (2024)[269]
Manfidokimab	Anti-IL-4Rα	III	NCT06383468 (2024)[270]
Rezpegaldesleukin	Anti-IL-2R	III	NCT06136741 (2026)[271]
Temtokibart	Anti-IL-22RA1	III	NCT05923099 (2026)[272]
Zumilokibart	Anti-IL-13	II	NCT07003425 (2026)[273]
Afimkibart	Anti-TL1A	II	NCT07223697 (2026)[274], NCT06863961 (2026)[275]
Barzolvolimab	Anti-KIT	II	NCT06727552 (2026)[276]
Camoteskimab	Anti-IL-18	II	NCT06436183 (2025)[277]
EVO301	Anti-IL-18	II	NCT06723405 (2025)[278]
FB825	Anti-CεmX	II	NCT05059509 (2022)[279]
APG990	Anti-OX40L	I	NCT07027527 (2026)[280]
ATTO-1310	Anti-IL-31	I	NCT06787586 (2026)[281]
BMS-986326	Anti-IL-2RA	I	NCT06248814 (2025)[282]
Multispecific antibodies			
Galvokimig	Anti-IL-13–IL-17A–IL-17F multispecific	III	NCT07277660 (2026)[283]
CM512	Anti IL-13–TSLP bispecific	II	NCT07230483 (2025)[284]
Lunsekimig	Anti-TSLP–IL-13 bispecific	II	NCT06790121 (2026)[285]
Ompekimig	Anti-IL-4–IL-31R–IL-33 multispecific	II	NCT05995964 (2026)[286]
Tilrekimig	Anti-IL-4–IL-31R–TSLP multispecific	II	NCT05995964 (2026)[286]
BBT001	Anti-IL-4Rα–IL-31 bispecific	I	NCT06808477 (2025)[287]
APG279 (APG777+APG990)	Anti-IL-13+ anti-OX40L bispecific	I	NCT07027527 (2026)[280]
Donzakimig	Anti-IL-13–IL-22 bispecific	I	NCT06716879 (2026)[288]
Small molecules			
Abrocitinib	JAK1 inhibitor	Regulatory approval	NA
Baricitinib	JAK1–JAK2 inhibitor	Regulatory approval	NA
Upadacitinib	JAK1 inhibitor	Regulatory approval	NA
Gecacitinib	JAK1–JAK2–JAK3+TYK2 inhibitor	III	NCT04539639 (2023)[289]
KT-621	STAT6 degrader	II	NCT07217015 (2026)[290]
Soquelitinib	ITK inhibitor	I	NCT07441395 (2026)[291]
SYX-5219	PKM2 modulation	I	ISRCTN80838388 (2025)[292]
Advanced topicals			
Crisaborole	PDE4 inhibitor	Regulatory approval	NA
Delgocitinib	JAK1–JAK2–JAK3+TYK2 inhibitor	Regulatory approval	NA
Difamilast	PDE4 inhibitor	Regulatory approval	NA
Roflumilast	PDE4 inhibitor	Regulatory approval	NA
Ruxolitinib	JAK1–JAK2 inhibitor	Regulatory approval	NA
Tapinarof	AhR agonist	Regulatory approval	NA
Asivatrep	TRPV1 agonist	III	NCT02748993 (2018)[293]
ATI-1777	JAK1–JAK3 inhibitor	II	NCT05432596 (2024)[294]
AMTX-100 CF3	Nuclear transport checkpoint inhibitor	II	NCT04313400 (2025)[295]
SHR0302	JAK1 inhibitor	II	NCT04717310 (2023)[296]

AhR, aryl hydrocarbon receptor; NA, not applicable; PDE4, phosphodiesterase 4. [a]Regulatory approval means approved by at least one regulatory body.

improvement of QoL in adults, and additional enhancement in sleep and depression–anxiety-related metrics[42].

The need for standardization of measurement and outcomes in AD led to the creation of HOME[10,41,43,53,249]. This initiative established a core outcome set with outcome measures to be reported in clinical trials. QoL was included as one of the four outcome domains, along with signs, symptoms and eczema control (Box 2). The HOME Core Outcome Set should be implemented in all clinical trials for systemic treatment in AD, endorsing AD interventions[249].

Outlook

The past decade has seen unparalleled advances in basic scientific insights, translational medicine and therapeutic options for people living with AD. Today's landscape would have been unimaginable to an informed observer in the mid-2010s. Despite these advances there remains much work to do across several domains. Here we posit our thoughts on the outlook for the next decade, identifying remaining challenges and potential opportunities.

Increasing our collective knowledge on the global burden of AD and its severity is crucial for healthcare planning and government and non-governmental organization involvement, as well as patient care. Considering this, the member states of the World Health Assembly adopted the 'Skin Diseases as a Global Public Health Priority' resolution to prioritize skin health from a global public health perspective,

including AD. From a research standpoint, the standardization of diagnostic criteria and epidemiological methods for skin diseases is a critical step towards this goal[34]. The International Study of Asthma and Allergies in Childhood has contributed substantially by developing a validated questionnaire to assess the prevalence of AD (as well as respiratory allergies) in paediatric populations, with more than two million children recruited globally[242]. There is also a need to develop internationally agreed definitions of what disease control means to patients and their healthcare professionals, More data need to be generated from epidemiological studies, especially in middle-resource and low-resource settings, in clinical trials with long-term follow-up and in disease registers.

There needs to be better equity of access to the novel life-transforming therapies including biologics, small molecules and non-glucocorticoid topical therapies at a global level. In 2025, the WHO added moisturizers to the Essential Medicines List for adults and children, a step that should improve basic access in low-income and middle-income countries; in dermatology more broadly, adalimumab and ustekinumab for psoriasis were also added to the Essential Medicines List, underscoring a path for advanced therapy access via global policy and biosimilars. Loss of patent protection and the rise of biosimilars will be a major catalyst for expanded access, although these will still probably remain inaccessible in low-income and middle-income countries.

Primary prevention of AD has been the focus of much research, especially since the implication of skin barrier defects through the discovery of *FLG* LoF mutations in 2006 (ref. 52). To date, early intervention trials, mostly focused on emollient applications in early life in study populations at varying levels of risk and with a wide range of moisturizers and treatment durations, have yielded heterogeneous and inconsistent results. Large pragmatic RCTs in high-risk infants or infants in the general population[250,251] found no preventive effect of daily emollients, whereas smaller, targeted studies[252] and a recent RCT in the community setting found modest risk reductions[252], highlighting how regimen and population are likely to drive outcomes. No well-replicated protocol can yet be recommended in practice.

The prospect of disease modification, currently a loosely defined term, which can refer to either permanent remission of the disease and/or, taking a more holistic approach, a decreased risk of developing atopic comorbidities over a person's lifetime, is a major therapeutic goal[253]. While some real-world evidence suggests that earlier and more aggressive treatment in early life may attenuate disease severity and lessen comorbidity burden, prospective, mechanism-anchored trials are still needed.

Compared with the most efficacious drugs approved for the treatment of psoriasis, another common immune-mediated inflammatory skin disease, all current AD therapies show lower efficacy (lower proportions of patients) in achieving clear or almost clear skin. In phase III trials, the absolute proportions of participants achieving an improvement in EASI of 75% from baseline (EASI-75) ranges from 73–80% for the highest dose of upadacitinib to 25–35% for the approved dose of tralokinumab[198]. A substantial fraction of patients only experience a partial or no response to these therapies, a much higher proportion than in psoriasis. This therapeutic efficacy gap remains an unmet need that may be closed by biomarkers that can guide therapy choice and switching in routine care, and by therapies in development with different modes of action.

The AD treatment landscape remains active (Table 1) with multiple phase II–III programmes under way[254]. Current strategies include antibodies against established targets with extended half-lives; for example, the long-acting anti-IL-13 zumilokibart for which phase II data suggest clinical activity, and extended dosing intervals are currently being explored[255], monoclonal antibodies against additional cytokine targets such as temtokibart, which blocks the IL-22RA1 receptor subunit[256] and multi-target biologics designed to co-block downstream effector cytokines[257]. Several agents targeting upstream T cell pathways, including OX40–OX40L modulators, are also in development[76]. Phase IIb trials of rocatinlimab and amlitelimab showed clinically meaningful improvements with signals of durability in some patients. However, in March 2026, the late-stage rocatinlimab programme was discontinued after additional cases of Kaposi sarcoma were observed and a mechanistic link to OX40 pathway modulation could not be excluded[258]. Although causality remains unproven, Kaposi sarcoma is an human herpesvirus 8-associated malignancy that is classically promoted by impaired immune surveillance[259], making vigilant long-term malignancy monitoring essential for upstream T cell-modulating strategies.

Beyond antibodies, early clinical programmes include targeted protein degraders (for example, an IRAK4 degrader and a STAT6 degrader (NCT06945458))[260], and regulatory T cell-directed approaches such as rezpegaldesleukin[261]. Bispecific and multispecific antibodies are the focus of much clinical development, and have thus far encouraging data have been reported for the IL-13–IL-17A and IL-17F multispecific galvokimig[262] and the IL-4–IL-13–TSLP multispecific tilrekimig[263], although an IL-4–IL-13–IL-31 multispecific programme has been dropped[264].

In conclusion, therapeutic options for AD have expanded substantially over the past decade, and a growing number of targeted and mechanism-informed approaches are in clinical development. At the same time, many investigational agents are supported primarily by early-phase studies, and their long-term effectiveness, safety and positioning in care pathways will need to be established in confirmatory phase III trials and real-world use.

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Competing interests
S.W. has received consulting fees, honoraria, grant support and/or lecture fees from AbbVie, Almirall, Apogee, Astria, Boehringer, Eli Lilly, Galderma, GSK, LEO Pharma, Pfizer, Regeneron Pharmaceuticals, Sanofi and UCB. V.A. has acted as an advisor for Galderma and Sanofi; and an investigator for AMGEN and Abbvie. N.C.D. received an honorarium or conducted clinal trials, and/or was a member of the advisory board, for Beiersdorf, L'oréal, Galderma, Piere Fabre, Sanofi and Aventis. S.D. was a member of the advisory board of Galderma, Sanofi and Cipla Immunology, with honoraria. K.K. has received consulting fees, honoraria, grant support and/or lecture fees from AbbVie, Amgen, Eli Lilly, Kyowa Kirin, Japan Tobacco, LEO Pharma, Maruho, Mitsubishi Tanabe, Ono Pharmaceutical, Pfizer, Procter & Gamble, Sanofi, Regeneron Pharmaceuticals, Taiho and Torii Pharmaceutical. L.M. has received research support and fees for consulting or as a speaker on atopic dermatitis from Regeneron Pharmaceuticals, Sanofi, Pfizer, Galderma, Leo Pharma, Janssen, Abbvie, Batanee and thederma, and declares no patents, ownership or financial gain from any atopic dermatitis drug. C.F. has received consulting fees, honoraria, grant support, and/or lecture fees from AbbVie, Almirall, Amgen, Apogee, Bioderma, LEO Pharma, Medac, Pfizer, Sanofi and Regeneron Pharmaceuticals. A.D.I. has acted as a consultant for AbbVie, Arena Pharmaceuticals, Dermavant, Eli Lilly, LEO Pharma, Menlo Therapeutics, Novartis, Pfizer, Regeneron Pharmaceuticals, Sanofi, Sitryx and UCB; has received speaker fees from AbbVie, Eli Lilly, LEO Pharma, Novartis, Pfizer, Regeneron Pharmaceuticals and Sanofi; has acted as an investigator for AbbVie, Novartis and Sanofi-Regeneron Pharmaceuticals and Sanofi; and is a patent holder with Regeneron Pharmaceuticals.

Informed consent
The authors affirm that human research participants provided informed consent for publication of the images in Fig. 5.

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